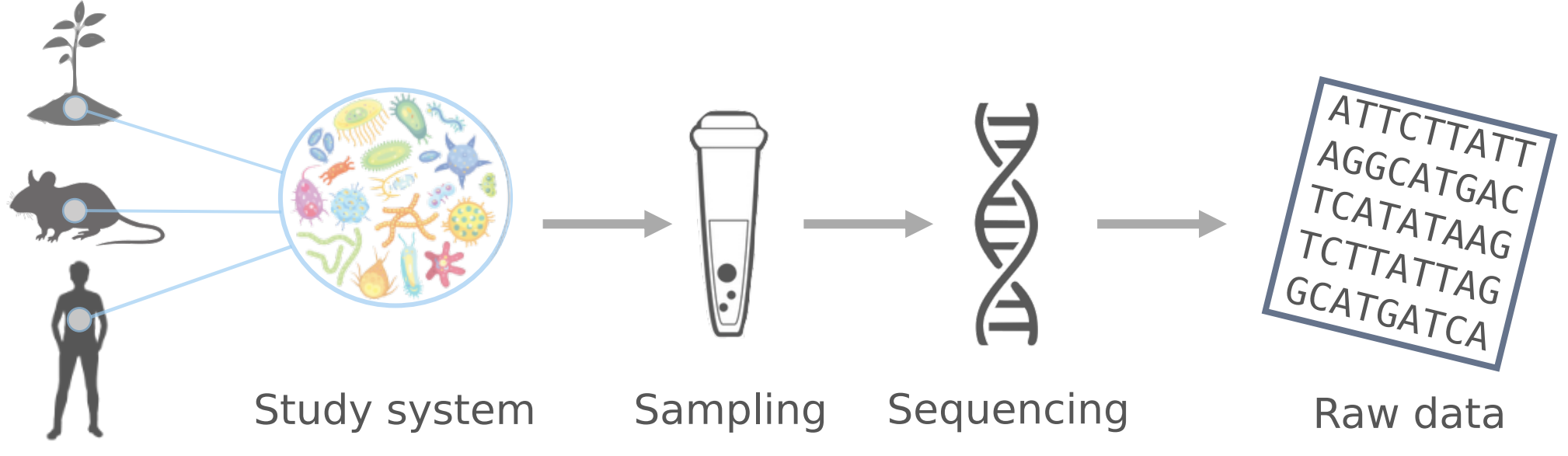
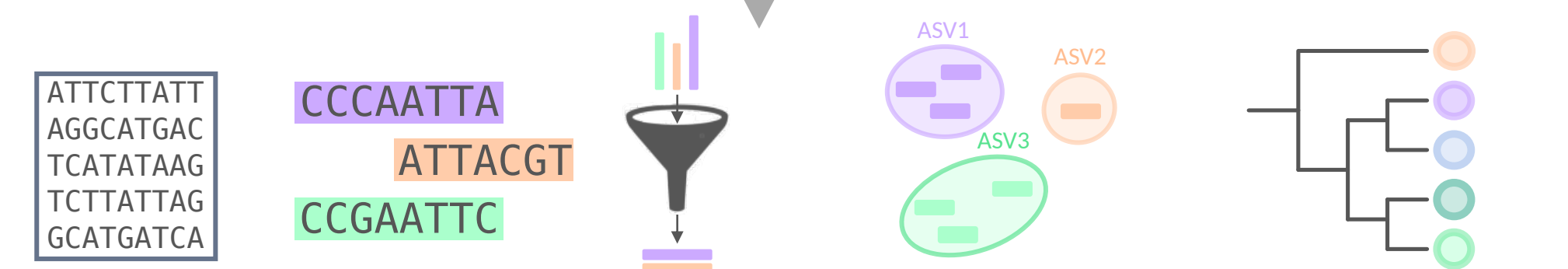
DATA GENERATION



Study system | Sampling | Sequencing | Raw data

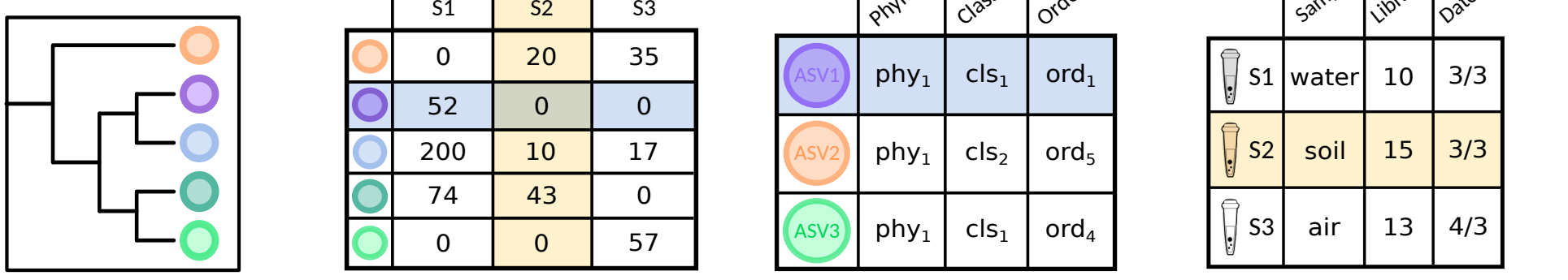
PREPROCESSING



FASTQ/FASTA files | Read alignment | Trimming/ Filtering | Unit clustering | Taxonomy/Functional tree construction

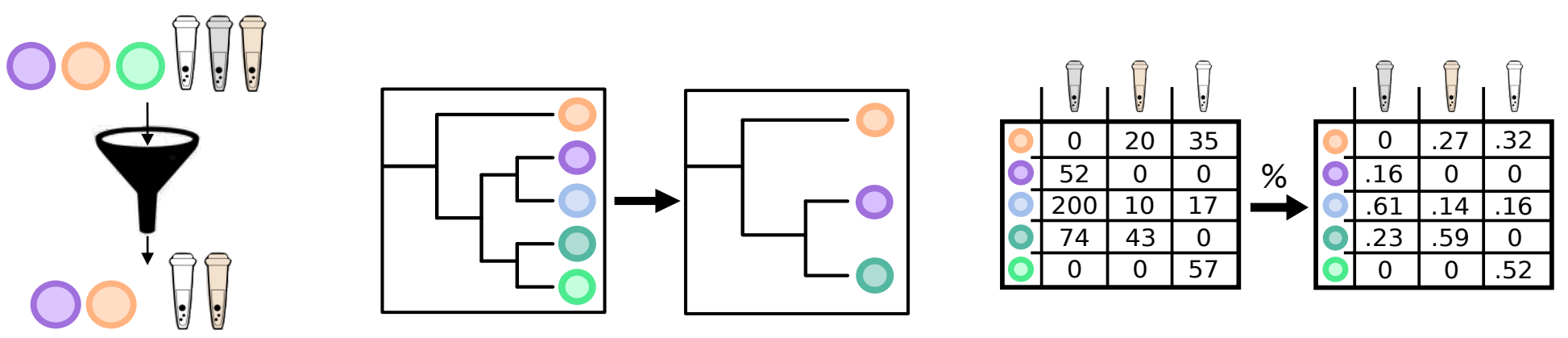
1. DATA OBJECT

TreeSummarizedExperiment



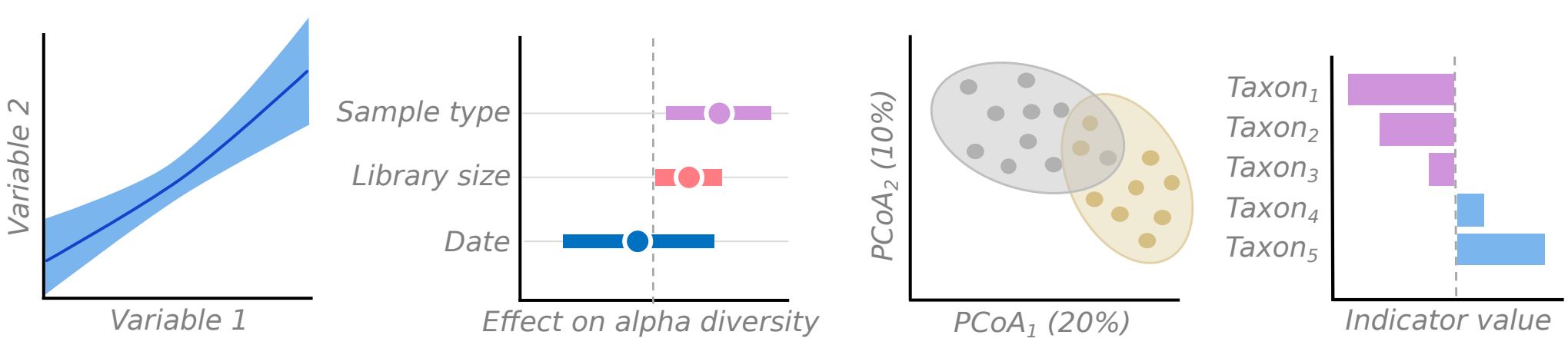
Tree | Abundance table | Taxonomy | Sample metadata

2. DATA PROCESSING



Quality control | Agglomeration | Transformation

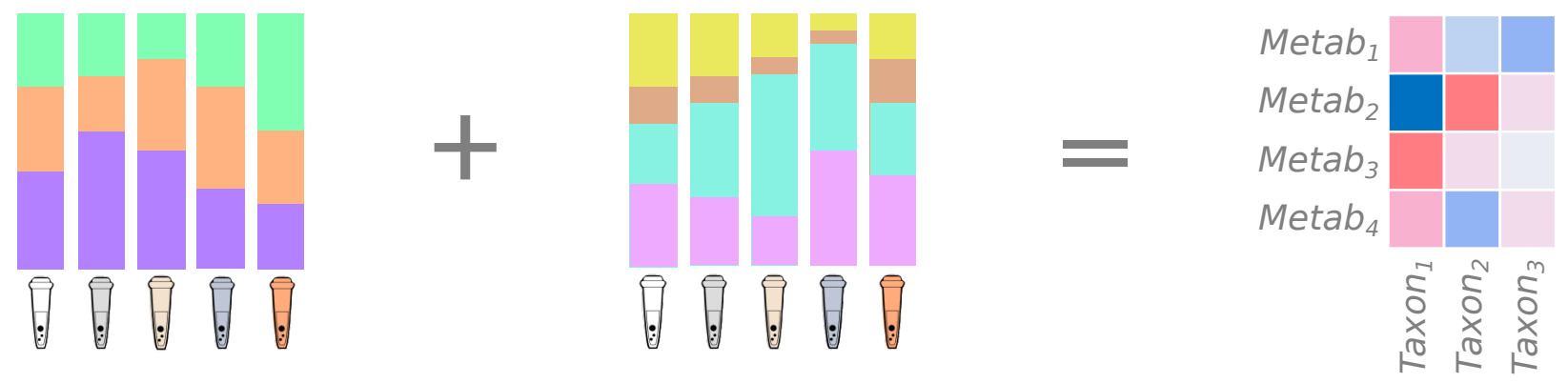
3. STATISTICAL ANALYSIS



Descriptive analysis | Alpha diversity | Beta diversity | Differential analysis

4. DATA INTEGRATION

MultiAssayExperiment



Taxonomic profiles | Metabolic profiles | Joint analysis

5. RESULTS

Reproducible reporting (*e.g.*, with Quarto)

Share your results in an open and reproducible way!

OMA